**D 11677** **(Pages : 3)** **Name.....**

**Reg. No.....**

**THIRD SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, NOVEMBER 2021**

[November 2020 for SDE/Private Students]

(CBCSS)

Mathematics

MTH 3C 13—FUNCTIONAL ANALYSIS

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

**General Instructions (Not applicable to SDE/Private Students)**

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section / Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section / sub part / sub division may be ignored.*
4. *There will be an overall ceiling for each Section / Part that is equivalent to the maximum weightage of the Section / Part.*

**Part A**

*Answer* **all** *questions.*
*Each question carries weightage* 1.

1. Consider the subspace $E_1$ of a linear space E. Prove that the dimension of $E/E_1$ is $n$ if and only if there exists linearly independent vectors $x_1, x_2 \ldots x_n$ linearly independent vectors relative to $E_1$ such that every vector of E can be uniquely expressed as a sum of their linear combination and a unique vector $y \in E_1$.
2. Is C [0, 1] a normed space ? Justify your answer.
3. State Holder's inequality and derive Cauchy Schwartz inequality from the same.
4. Show that inner product $\langle x, y \rangle$ is a continuous function with respect to both variables .
5. Prove that any two separable infinite dimensional Hilbert spaces $H_1$, $H_2$ are isometrically equivalent.
6. Let $f \in E^H / \{0\}$. Show that *codim ker f* = 1.

**Turn over**

7. State Arzela's theorem.

8. Show that there exists a linear functional on a normed space X that distinguishes distinct elements of X.

(8 × 1 = 8 weightage)

## Part B

*Answer* **six** *questions choosing* **two** *from each unit.*
*Each question carries weightage* 2.

UNIT 1

9. Show that norm is a continuous function.

10. Is a quotient space a normed space ? Justify your answer.

11. Show that the kernel for a seminorm $p$ is a subspace of a linear space on which it is defined. Also show that $p(x + y)$ is independent of, where $y$ is an element of the subspace.

UNIT 2

12. State Bessel's inequality and use it to show that a complete orthonormal system in a Hilbert space H is a basis in H.

13. Prove that $f$ is a bounded functional on a normed space X if and only if $f$ is continuous.

14. If E is a closed subspace of a Hilbert space H and codim E = 1 then prove that $E^{\perp}$ is a one dimensional subspace.

UNIT 3

15. Show that $l_1$ can be identified as the dual space of $c_0$.

16. Prove that the dual space of any normed space is complete.

17. Let X, Y be any *two* Banach spaces .Prove that for a linear operator $A : X \to Y$ implies $A^* : X^* \to Y^*$ is compact.

(6 × 2 = 12 weightage)

## Part C

*Answer* **two** *questions.*
*Each question carries weightage* 5.

18. Let E be a normed space. Show that there exists a complete normed space $\hat{E}$ and linear operator $T : E \to \hat{E}$ such that :

    (i) $\|T(x)\| = \|x\|$.

    (ii) *Im* (T) is a dense set in $\hat{E}$.

19. State and prove a necessary condition for a Hilbert space to have an orthonormal basis.

20. (a) Consider $f \in E^* / \{0\}$. Prove that :

    (i) *codim kerf* = 1.

    (ii) $f, g \in E^{\#} / \{0\}$ and ker $f$ = ker $g$ *then there exists* $\lambda \neq 0$ such that $\lambda f = g$.

    (iii) If L is a closed subspace of E and *codim* L = 1 then there exists $f \in E^{\#}$ such that ker $f$ = 1.

    (b) Illustrate with an example the concept of non-separable Hilbert space.

21. (a) Discuss the compactness of the integral operator in $L_2$.

    (b) State and prove necessary and sufficient condition for a set to be relatively compact in a normed space.

(2 × 5 = 10 weightage)